Negative Literature Answer to the RN Continuous-Image Problem in arXiv:1005.4303

FA/Banach run packet

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Status

This is a literature-already-answered packet. It records that the continuous image problem for Radon–Nikodym compact spaces, stated in the selected-problems survey of Aviles–Kalenda [1], was answered negatively by the later paper of Aviles–Koszmider [2]. It is not an original proof from this run.

Original Problem

In Section 6, "Radon-Nikodym compact spaces", Aviles–Kalenda write that the main open question since Namioka is whether continuous images of RN compacta are again RN compact. They then ask:

> Is the continuous image of an RN compact space also RN compact?

The local source location inspected was `data/parsed/arxiv_sources/1005.4303/probl5.tex`, lines 627–633.

Supporting Answer

The supporting paper [2] is titled *A continuous image of a Radon-Nikodym compact which is not Radon-Nikodym*. Its abstract states that the authors construct a continuous image of an RN compact space which is not RN compact, solving Namioka's problem. The precise theorem used here is Theorem 1.1:

> There exists a continuous surjection $\pi : L_0 \to L_1$ such that L_0 is a zero-dimensional RN compact space but L_1 is not RN compact.

Literature Answer. *The continuous-image question from Section 6 of [1] has a negative answer. A continuous image of an RN compact space need not be RN compact.*

Literature identification. Theorem 1.1 of [2] supplies a compact space L_0, a compact space L_1, and a continuous surjection $\pi : L_0 \to L_1$. The domain L_0 is RN compact, indeed zero-dimensional RN compact, while the image L_1 is not RN compact. Thus L_1 is a continuous image of an RN compactum but is not RN compact. This is exactly the negation of the question asked in [1].

The supporting authors also explicitly identify the construction as a solution of the problem posed by Isaac Namioka, so this is an explicit later answer rather than an agent-identified implication. □

Scope Limitations

This packet clears only the headline continuous-image problem at the start of Section 6 of [1]. It does not resolve the nearby questions in that survey about quasi-RN compacta, cardinal reductions, the space of almost increasing functions, convex hulls of RN compacta, zero-dimensional RN covers, embeddings into measure spaces over scattered compacta, or unions of two RN compacta.

Verification Notes

The source and supporting papers are distinct. The supporting paper appeared in 2011, after the 2010 selected-problems survey, and explicitly says that it solves Namioka's problem. The local duplicate search covered arXiv ids `1005.4303` and `1112.4152`, the survey title, and phrases `continuous image of an RN compact`, `Radon-Nikodym compact`, and `Namioka`.

References

[1] A. Aviles and O. F. K. Kalenda, *Compactness in Banach space theory - selected problems*, arXiv:1005.4303.

[2] A. Aviles and P. Koszmider, *A continuous image of a Radon-Nikodym compact which is not Radon-Nikodym*, arXiv:1112.4152.